Can one local reading pass Cassini and keep a wide-binary signal? The QUMOND chain of MM 2.56 at 17σ, and a reading registered for Gaia DR4

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Abstract

Cassini ranging data bound the external-field quadrupole of the Solar System at $Q_2 = (1.6 \pm 1.8)\cdot 10^{-27}$ s^{-2} (Park et al. 2026). The wide-binary chain registered in MM 2.56 has these ingredients: Taken down to the Sun with the same field equation, it predicts $Q_2 = 3.22\cdot 10^{-26}$ s^{-2}, 17σ above the measurement (17.0–19.4σ over eight input choices; the MM 2.56 solver gives the same number within $3\cdot 10^{-5}$). A local reading of the external field. Here the anomaly lives in the relation between two bodies. What it gives. Registered for Gaia DR4. The reading has its own orbit library, the unchanged forward model of MM 2.56, and a decision rule against Newton (threshold ±5 in log-likelihood, subsets ±2). The copied evaluation reproduces the MM 2.56 dry run exactly. A draft protocol with three addenda registers the anisotropy test:

Version 2.60 of the Mathematical Materiality (MM) series, October 2026.

- A result record, with a reading of the external field registered for Gaia DR4 (planned for 2 December 2026).
- It continues MM 2.56 (DOI 10.5281/zenodo.23506401), which registered a wide-binary test for DR4. MM 2.56 and its protocol are unchanged.
- The reference texts are in Italian, in documenti/; this README summarises them.
- In the working records this deposit is called 2.58.

In brief

- **The measurement.** Cassini radio tracking bounds the external-field quadrupole of the Solar System: $Q_2 = (1.6 \pm 1.8)\cdot 10^{-27}$ s^{-2} (Park et al. 2026, PRD; arXiv:2602.17884).
- **The QUMOND chain at the Sun.** The chain of MM 2.56 uses:
- $a_0 = cH_0/6$;
- the law ν(y) = 1/(1 − e^(−√y));
- the Milky-Way field V²/R;
- QUMOND.

Taken down to the Sun it predicts $Q_2 = 3.22\cdot 10^{-26}$ s^{-2}, 17σ above the measurement (17.0–19.4σ over eight input choices). Cassini excludes the global-field reading down to the Sun.

- **The local reading** (section 4). The anomaly is read in the relation between two bodies:
- the body whose local MCD prevails reads the external field for its system;
- a dominant compact body reads the relation itself;
- a relation without a dominant body is read along its orbit;
- interior and exterior fields add as magnitudes;
- the direction enters through a tensorial alignment, with an amount measured from the data;
- light reads the curvature and not the external field.
- **What it gives:**
- Saturn: $Q_2 \approx 10^{-30}$ s^{-2};
- planets and comets with any orbit: Newtonian; the star S2 around Sgr A*: no anomalous boost (general relativity remains);
- the 919 low-acceleration wide binaries of MM 2.56: a boost $\langle\eta\rangle = 1.27$, against 1.32 for the QUMOND chain and 1 for Newton;
- disc galaxies: the radial acceleration relation unchanged. In 78 SPARC galaxies the environment field enters in full, as in the binaries;
- weak lensing (KiDS-1000): $\chi^2 = 219.9$ on 60 points, against 1 687.8 for Newton;
- around galaxies in denser environments, light reads two to three times more at large radii (all lenses against isolated ones). Among isolated central galaxies, red and blue differ by a nearly constant ratio, so the excess is

compatible with the satellites of the full sample rather than with a reading of light that grows with density.
- **Systems without rotation** (section 4.9):
- classical dwarf spheroidals agree within ±0.15 dex;
- ultra-faint dwarfs move 2.5–20 times faster than the reading predicts;
- distant globular clusters are Newtonian.

At the same gravitational state (x_h + e_N 0.019–0.12: 19 dwarfs, 11 clusters), galaxies carry the boost (+1.0 dex over Newton) and clusters do not (+0.02).

- The systems whose stars belong to different generations (metallicity spread) carry a mirror; single-generation clusters do not.
- The amount follows a common profile set by the radius, M_dyn ∝ r_½^1.54±0.11, for satellites and isolated dwarfs alike, with an intrinsic scatter of 0.48 dex.
- **Registered for DR4** (section 5):
- the boost, with its own orbit library, the forward model of MM 2.56 and a decision rule against Newton;
- the anisotropy, in a draft protocol with an addendum.
- **On DR3** (data already seen), in the registered simulation series the local reading has the highest likelihood of the seven hypotheses: +16.1 in log-likelihood over Newton, +6.1 over MOND simple, +8.5 over the QUMOND chain.
- **Measured simulation noise** (section 5.3). Over four series with new seeds and new libraries:
- the local reading over Newton is +11.0 on average (from +8.2 to +16.1), above the threshold +5 in every series;
- the QUMOND chain over Newton ranges from −0.9 to +12.1;
- the local reading over the chain ranges from −2.0 to +9.0.

A rule over several series is registered for DR4.

1. What is known and what is new

- **Known.**
- The external-field quadrupole in modified-Poisson MOND: Milgrom 2009; Blanchet and Novak 2011.
- The Cassini constraints: Hees et al. 2014 and 2016; Park et al. 2026.
- Within QUMOND/AQUAL, Cassini implies a Newtonian wide-binary test: Desmond, Hees and Famaey 2024; Desmond 2025; Park et al. 2026.
- Long-period comets disfavour AQUAL: Vokrouhlický, Nesvorný and Tremaine 2024.
- Orbit-dependent (modified-inertia) readings: Milgrom.
- The external-field effect in SPARC rotation curves: Chae et al. 2020, 2021.
- The radial acceleration relation in weak lensing: Brouwer et al. 2021.
- **New here.**
- The chain's number, with no input taken from the binaries, reproduced by the MM 2.56 solver within $3\cdot10^{-5}$.
- A pair-by-pair location of the 919 pairs in the region where the quadrupole is generated.
- One local reading that holds together:
- Saturn, comets and S2;
- wide binaries;
- disc galaxies in their environment;
- weak lensing.
- A check of dwarf galaxies and distant globular clusters with the same reading. They split: galaxies boosted, clusters Newtonian. Who carries the mirror (stars of different generations) and how much (a common profile set by the radius) are measured.

Its DR4 profile is registered.

- A dipole statistic for wide binaries that the sample systematics do not reach.

2. Data (documenti/INVENTARIO_LIVELLO0_v0.md and _v1_AGGIUNTA.md; dati/FONTI.md)

quantity	value	source
Q_2 ($\delta\Phi = -(Q_2/2)[(\hat{e}\cdot x)^2 - r^2/3]$, $\hat{e}$ toward the Galactic centre)	$(1.6 \pm 1.8)\cdot10^{-27}$ s^{-2}	Park et al. 2026, eq. 6, table 1
Solar-System acceleration	$(2.32 \pm 0.16)\cdot10^{-10}$ m s^{-2}	Gaia EDR3, Klioner et al. 2021

Milky-Way field of the chain	$V^2/R = 2.1456\cdot10^{-10}$ m s^{-2} (V = 233 km/s, R = 8.2 kpc)	MM 2.56
$a_0 = cH_0/6$	$1.0914\cdot10^{-10}$ (H_0 = 67.4); $1.1827\cdot10^{-10}$ (H_0 = 73.04) m s^{-2}	MM 2.54, MM 2.56
Baryonic surface density	Σ_b = 47 M⊙ pc^{-2}	Park et al. 2026, after Bland-Hawthorn and Gerhard 2016
Pairs	919 low-acceleration (s > 5000 au, x < 1), 21 471 control	MM 2.56, dati/campione_F.csv (El-Badry, Rix and Heintz 2021)
Eccentricity law of the pairs	p(e) ∝ e^α, α(s) of Hwang et al. 2022	as in MM 2.56
Supplementary perihelion precessions	EPM2011 (Pitjeva and Pitjev 2013); INPOP10a uncertainties (Fienga and Minazzoli 2024)	—
Light deflection	$\gamma - 1 = (2.1 \pm 2.3)\cdot10^{-5}$	Bertotti, Iess and Tortora 2003 (via Will 2014)
Planetary elements	J2000, table 1	JPL approximate positions (dati/approx_pos.html)
Weak lensing	KiDS-1000 lensing rotation curves, 4 mass bins, 60 points, full covariance	Brouwer et al. 2021 (as in MM 2.44)
Rotation curves	SPARC table 1 and 2	Lelli, McGaugh and Schombert 2016
Environment field of SPARC galaxies	log e_N,env, table 3	Chae et al. 2021 (arXiv:2109.04745)
Wide binaries for the direction	57 583 pairs beyond 5 000 au; the catalogue beyond 300 au	El-Badry, Rix and Heintz 2021 (VizieR J/MNRAS/506/2269), queried in August 2026
Dwarf galaxies of the Local Group	σ★, V, E(B−V), r_h, ellipticity, distance modulus	McConnachie 2012, version of January 2021 (CADC)
Globular clusters	σ_0, mass and M/L_V (for L_V), r_h, R_GC	Baumgardt database, combined table of January 2026
Dwarfs and clusters with metallicity spreads	σ_v, σ[Fe/H], r_½, M_V, distances, host	Local Volume Database (Pace 2025), comb_all.csv of 12 August 2026

The papers and the larger data files are not included; dati/FONTI.md lists them with URLs and SHA-256.

3. The QUMOND chain taken to the Sun (documenti/PROTOCOLLO_LIVELLO1.md, NOTA_ESITO_LIVELLO1.md)

Outcome, in the sentence written before the computation: the chain of MM 2.56, taken to the Sun with the same field equation, is not compatible with Cassini ($Q_2 = 3.22\cdot10^{-26}$ s^{-2}, z = 17.0).

H_0	external field	conversion to Newtonian	e_N	Q_2 (10^{-26} s^{-2})	z
67.4	V^2/R	spherical (**main case**)	1.351	**3.221**	**17.0**
67.4	V^2/R	with disc	1.369	3.254	17.2
67.4	Gaia	spherical	1.501	3.499	18.6
67.4	Gaia	with disc	1.517	3.526	18.7

73.04	V^2/R	spherical	1.210	3.328	17.6
73.04	V^2/R	with disc	1.228	3.367	17.8
73.04	Gaia	spherical	1.347	3.625	19.3
73.04	Gaia	with disc	1.362	3.658	19.4

- **Method.** The phantom-density moment. It reproduces Park's 3.387 and $3.411 \cdot 10^{-26}$ exactly; the MM 2.56 solver gives the same Q_2 within $2.8 \cdot 10^{-5}$. z uses the Cassini error.
- **The boost at the Sun.** The chain asks for a radial boost of about 1.45 at the Sun, where Park et al. allow at most 1.010–1.018 within QUMOND.
- **Where the signal is generated** (r_M of the Sun = 7 371 au):

r/r_M	share of Q_2	low-band pairs	control pairs
< 0.30	8.2 %	0	21 461
0.30–0.95	61.2 %	0	10
0.95–1.93	24.2 %	799	0
> 1.93	6.4 %	120	0

In the scale s/r_M the low-band pairs sample the outer part of the region where the quadrupole is generated.

- **Precessions.** The same Q_2 gives Saturn a perihelion drift of +4.5 mas/century, against -0.32 ± 0.47 in EPM2011.
- **Light deflection.** The metric sector gives $\gamma = 1$ by structure, as measured.

4. The local reading (documenti/STRUTTURA_LETTURA_LOCALE.md; numbers in documenti/CONTROLLI_DELLA_STRUTTURA.md)

4.1 Where the anomaly lives

- In QUMOND the anomaly is a phantom density in the empty space around a mass. Around the Sun it forms a cloud thousands of au wide, which generates the quadrupole excluded by Cassini.
- In the local reading the anomaly lives in the relation between two material bodies.
- The phantom cloud is a tool of the QUMOND equation, not an object.

4.2 Who reads the external field: the prevailing local MCD

- **The rule.** The local MCD of a mass is R_s/r.
- In a relation, the MCD of each body is taken at the place of the other, that is at the same distance r. Their ratio is therefore the mass ratio.
- The body whose MCD prevails reads the external field for its whole system.
- **Solar System.** The Sun's MCD at a planet exceeds the planet's MCD at the Sun by the mass ratio: 1 048 for Jupiter, 3 499 for Saturn, more for the others, and about 10^{17} for a comet.
- Planets and comets read the Galaxy through the Sun, so the Sun and Saturn receive the same Galactic acceleration.
- Without this rule they would differ by $6.7 \cdot 10^{-11}$ m s^{-2}. As a scale estimate, $\delta a/n^2 \approx 1\,500$ km for Saturn's orbit, against radio tracking at the level of metres.
- **Galaxies.** A galaxy prevails over its stars and gas by mass ratios of 10^6–10^{11}. They read the field of the environment through the galaxy.
- **Wide binaries.** The pairs have mass ratios of at most 7.4, so each star reads the Galaxy itself.
- **Threshold.** «Prevailing» is any threshold between 7.4 and 1 048. No number has to be tuned.

4.3 A dominant compact body reads the relation itself

- **The rule.** When a body prevails and is compact with respect to the relation (radius R_dom much smaller than

the distance r), the relation is read with the MCD of the dominant body at its own place, its surface. There the reading is very high ($x = 2.5 \cdot 10^{12}$ for the Sun), so the relation is Newtonian.
- **An extended dominant body** (a galaxy around its stars and gas) is spread along the relation, so the reading stays local.
- **Threshold.** The compactness threshold R_dom/r is any value between $7 \cdot 10^{-4}$ (Sgr A*–S2 at pericentre, the least compact of the compact cases) and $2.5 \cdot 10^{-2}$ (the Milky Way with a satellite at 100 kpc, the most compact of the extended cases).
- **What it gives:**
- planets and comets on any orbit, including new Oort-spike comets that never came close to the Sun: Newtonian;
- S2: no anomalous boost, so general relativity remains, as measured;
- wide binaries: no star prevails, so the boost remains;
- galaxies: the radial acceleration relation remains, also for outer gas and satellites.

4.4 A relation without a dominant body: the inertial state

- **The rule.** The reading of a relation is its time average along the orbit: $x_{eff} = \langle x \rangle = x(a)/\sqrt{(1 - e^2)}$, with $x = GM/(r^2 a_0)$.
- It applies to relations in which no body prevails, such as wide binaries. On DR3 the orbit average fits better than the instantaneous reading (section 5).
- On circular orbits $x_{eff} = x$, so the radial acceleration relation of disc galaxies is unchanged.
- **The DR3 gain sits in the fast pairs.** On DR3 the gain over Newton comes entirely from the fast pairs: the top third of the u classes gives +10.6 of about +11, averaged over four series (lettura_locale/coppie/centrifuga.py).
- A reading that made the fast phases Newtonian, where the centrifugal force prevails over the gravity read, would remove the boost exactly there. The data exclude it: the registered criterion gives −10.4 (s 3.2) instead of a positive value. The u classes alone do not say in which orbital phase a pair is.

4.5 How interior and exterior combine

- **The boost** of a relation is $\nu(x_{eff} + e_N)$, with:
- $a_0 = cH_0/6$;
- $e_N = 1.351$, the Newtonian Galactic field at the Sun, obtained from $V^2/R = 1.966\, a_0$ through ν.
- **The forms compared.** Among the registered forms (vector sum, quadrature, sum of magnitudes, relation without exterior), the sum of magnitudes gives the pairs, in the instantaneous form, a boost close to the chain's (+2 %). The others give 28–203 % more.
- **With the inertial state** the per-pair mean is $\langle \eta \rangle = 1.27$, an excess 15 % below the chain's.
- This estimate takes the observed separation as the semi-major axis.
- In the full forward comparison on DR3, with whole orbits, the reading fits better than the chain (section 5).
- **The external field read by matter is the baryonic one.** With the total field, dark-matter mirror included (e_N $\nu(e_N)$), the SPARC curves worsen by $\Delta\chi^2 = +2\,757$ and the dwarfs worsen too (lettura_locale/campo/specchio_campo.py).
- In the wide binaries the full forward model on DR3 does not separate the two fields. With the total field in place of e_N in the argument of ν (the reading under test) the likelihood changes by −1.4 against the mean of the 2.60 reading, within the registered thresholds of ±6.98 (lettura_locale/campo/valuta_totale.py). Against the QUMOND chain the difference is +2.5.
- **No significant dependence on the mass ratio** below the prevalence threshold. On DR3, after removing the sample effect seen in the close pairs, unequal pairs ($q < 0.5$) differ from similar ones by -0.03 ± 0.05 dex (lettura_locale/coppie/ rapporto_masse.py).
- **The value of the Galactic field.** With the Gaia field (about 8 % stronger than V^2/R) the mean boost of the 919 pairs goes from 1.2745 to 1.257, an excess 6–7 % smaller (lettura_locale/campo/sensibilita_campo.py).
- **The screening readings.** Charge reserve, crossing filter, 30 pc cell and history factor were also registered. They silence Saturn and make the binaries Newtonian too; this is the declared alternative of MM 2.56 (documenti/ PROTOCOLLO_LIVELLO2.md, NOTA_ESITO_LIVELLO2.md).

4.6 Direction: a tensorial alignment (lettura_locale/direzione)

- **The Galactic MCD difference across the pair.** The star nearer the Galactic centre reads a higher Galactic MCD.
- Along the line joining the two stars, the difference is $2\, g_e\, s \cos\theta/c^2$.
- The pair's own MCD is $2GM/(s\, c^2)$.
- Their ratio $q = (g_e/a_0)/x$ sets the weight of the direction. Its median is 3.2 for the 919 wide pairs and 0.018 for the close pairs.
- **The reading is a tensorial alignment.** The line of the pair contracts with the direction of the centre, $\hat{s}_i \hat{s}_j \hat{e}_i \hat{e}_j = \cos^2\theta$. More MCD read means more speed:

$v^2 = v_0^2 (1 + w\, q \cos^2\theta)$.

The boost is largest when the pair is aligned with the Galactic centre.

- **The amount w is an observational input**, measured and not derived.
- The data of August 2026 give w ≈ 0.007 (0.001–0.014 at 2σ) with a 3D statistic, and w = 0.024 ± 0.022 from the quadrupole on the sky.
- With w ≈ 0.007 the mean boost of the wide pairs changes by +0.002 dex, so section 4.5 is unaffected. The registered DR4 simulations do not include this term.
- The DR4 program, run on DR3 with the close-pair level subtracted and the full covariance (section 5.5), gives w = 0.005 ± 0.010 (symmetric form) and −0.003 ± 0.009 (dominant form): compatible with zero, with upper limits 0.024 and 0.014. The values above contained the sample level of the close pairs.
- **The form is symmetric.** When the companion lies behind the star, on the far side from the centre, the star in front still moves with the anomaly because of the companion. The boost is present on both sides, so on the sky there is only the cos 2φ term and the dipole is zero.
- **The alternative form,** also registered, is read by the dominant star as in section 4.3. The boost is present only when the secondary lies between the primary and the centre, which would give a dipole cos φ about 1.9 times the quadrupole.
- **Data of August 2026** (9 378 pairs at 5 000–50 000 au, region w > 0.5):
- dipole +0.0027 ± 0.0046 dex and quadrupole +0.0050 ± 0.0046 dex;
- close pairs (300–2 000 au): quadrupole +0.0041 ± 0.0018 dex, where the prediction is zero, and dipole +0.0001 ± 0.0018 dex;
- the Galactic north pole, used as a control direction, shows nothing.
- **The dipole is clean.** The Gaia scanning law and the selection do not know which star is the brighter with respect to the Galactic centre.

4.7 Light reads the curvature (lettura_locale/luce)

- **The rule.** Light reads the curvature of the MCD with the same law as the stars, without extra mass.
- **Light does not read the external field:** a uniform field has no curvature. Matter reads relations, and there the external field enters.
- **KiDS-1000 lensing** (Brouwer et al. 2021, 60 points, full covariance):
- $\chi^2 = 219.9$ with $a_0 = cH_0/6$, against 1 687.8 for Newton. The program reproduces the MM 2.44 values (236.61 and 1 687.81).
- An environment field added as for matter lowers the signal at 1–2.6 Mpc by factors of 9–17 and is excluded for all the values tried, 0.003 to 0.1 a_0 ($\Delta\chi^2 \geq 552$).
- As in MM 2.44, no model fits fully with the covariance provided ($\chi^2/60 \approx 3.7$).
- **Denser environments and central galaxies.**
- KiDS-1000 all lenses against isolated lenses (Brouwer et al. 2021, Fig-A4 and Fig-9): the ratio is 2.0–3.0 at the lowest g_bar and returns to 1 above $4 \cdot 10^{-14}$ m s^{-2}. The excess grows with R (slope −0.90 ± 0.14 in g_bar).
- Isolated lenses have no brighter galaxy within 3 Mpc, so they are not satellites of larger galaxies. Among them red and blue galaxies (and high and low Sérsic index) differ by a nearly constant ratio: slopes −0.28 ± 0.10 and +0.02 ± 0.09. A growth as R^2 (slope −1) is disfavoured ($\Delta\chi^2$ +16 and +21; lettura_locale/luce/lenti_rosse_blu.py). The errors treat the two samples as independent; the covariance from shared sources is not provided.
- The growing excess of the full sample is compatible with its satellites, for which the host enters the signal only beyond their distance from it. The red/blue comparison does not prove it, since the density difference of their environments is not measured. A distortion growing as R^2 does not appear among isolated lenses.

4.8 Galaxies in their environment (lettura_locale/galassie)

- **The rule.** The environment field enters in full and adds as a magnitude, as in the binaries.
- **78 SPARC galaxies** (1 419 points), with the environment field of Chae et al. 2021 galaxy by galaxy:

environment field	$\Sigma\chi^2$	$\Delta\chi^2$
in full, added as a magnitude	**12 490.8**	**0**
diluted by the ratio of the galaxy's mean MCD to a star's	13 341.2	+850.4
none	13 447.5	+956.7

- **The size of the effect.** The environment field has median $1.9 \cdot 10^{-3}$ a_0; at x = 0.01 it lowers g by 9 %.
- **Concentration.** Five declining curves give 865 of the 957 points. Without them the full field still leads by 92 (38 of the remaining 73 galaxies improve, 35 worsen; lettura_locale/galassie/cinque_curve.py).
- **Fit quality.** Distances and inclinations are fixed at the SPARC values, so χ^2 per point is about 9.

4.9 Systems without rotation: dwarf spheroidals and globular clusters (lettura_locale/sistemi)

- **The reading.** The stars do not rotate in order but move in all directions. A star reads the relation with the rest of the system along its line through the centre, with the increase of MCD set by the local state: ν(x_h + e_N).
- **The check.**
- Samples: 35 Milky-Way satellites at 20–300 kpc (McConnachie 2012, January 2021 version) and 25 globular clusters beyond 20 kpc (Baumgardt database, January 2026).
- Method: the Wolf et al. 2010 estimator with M_b = Y★ L_V (Y★ = 2 for dwarfs, 1.7 for clusters), and the Milky Way as $6 \cdot 10^{10}$ M☉ of baryons; nothing is fitted.

sample	Newton	ν(x_h + e_N)	ν(x_h), no external field	boost on one axis only
dwarfs (35)	+1.00 (χ^2 1 629)	+0.54 (χ^2 598)	+0.31 (χ^2 200)	+0.68 (χ^2 939)
clusters (25)	+0.05 (χ^2 27)	−0.16 (χ^2 163)	−0.29 (χ^2 255)	−0.04 (χ^2 64)

The entries are median log10(σ_obs/σ_pred), in dex.

- **Classical dwarfs** with ν(x_h + e_N): Fornax −0.08, Leo I −0.04, Leo II +0.05, Sculptor +0.11, Carina +0.15. Draco, Ursa Minor and Sextans: +0.40 to +0.42.
- **Ultra-faint dwarfs:** +0.4 to +1.3 dex. The residual grows with the weight of the Milky-Way field (Spearman ρ = +0.73).
- **Globular clusters** are Newtonian even at 100–150 kpc: Palomar 14 +0.06, NGC 2419 +0.03, Crater −0.04, AM 1 −0.13.
- **Same gravitational state.** With x_h + e_N in the range common to both samples (0.019–0.12):
- 19 dwarfs sit +0.995 dex above Newton (+0.690 with ν(x_h + e_N));
- 11 clusters sit +0.022 dex above Newton (−0.387 with ν(x_h + e_N)).
- **Stellar mass-to-light Y★ × 10^(±0.15)** leaves the conclusions unchanged:
- dwarfs with ν(x_h + e_N): +0.62 to +0.47;
- clusters: Newtonian in all three cases (lettura_locale/sistemi/sistemi_sensibilita.py).
- **Without the external field** the dwarfs fit better (median +0.31, χ^2 200 against 598) but stay above the prediction.
- **Who carries the mirror: systems with constituents of different generations.** The stars of a dwarf were born at different times, those of a cluster nearly together. In the reading, this distance in time between constituents gives the density, and so the MCD, on which the mirror depends. The data show the association; the metallicity spread is also one of the criteria astronomers use to separate galaxies from clusters.
- In the Local Volume Database all 38 systems with a measured metallicity spread carry a mirror (B = M_dyn/M_b > 3). Two of them are catalogued as clusters (Balbinot 1, DELVE 4).
- Clusters with nearly zero spread (limits 0.05–0.14 dex) are Newtonian.
- Within galaxies the amount does not follow the spread (ρ = +0.01).
- **How much: a common profile set by the radius.** The dwarf is an element on its own, with an autonomous mirror whose extent is given by its radius.
- For 81 dwarfs (Milky-Way and M31 satellites and isolated ones): log M_dyn = 7.064 + 1.53 (log r_½ − 2.5), that is $3 \cdot 10^5$ M☉ at 30 pc, $1.1 \cdot 10^7$ at 300 pc, $6.8 \cdot 10^7$ at 1 kpc.
- The exponent is 1.54 ± 0.11, with the error model in which the radius error enters M_dyn and r_½ together. Without the gas-rich systems (measured HI) it is 1.40 ± 0.11. In both cases 2 (constant acceleration) and 1 (constant velocity) are excluded and 1.5 is compatible.
- The amount grows much less than the visible mass and does not follow the metallicity spread. At fixed radius a weak dependence on the visible mass remains (r = +0.39 at fixed tide).
- Satellites and isolated dwarfs lie on the same profile. Without gas-rich systems, however, only three isolated dwarfs remain (Tucana, Cetus, Andromeda XVIII): difference −0.03 dex, interval −0.50 to +0.13.
- Around the profile an intrinsic scatter of 0.48 ± 0.10 dex remains.
- Part of it is associated with the Milky-Way tide: less mirror where the tide weighs more (r = −0.41, p = 0.02, at fixed visible mass).
- Part may come from binaries and non-equilibrium among the ultra-faints.
- At r_½ the dwarfs share an acceleration of about 0.17 a_0, of the order of the Milky-Way field at those distances (about 0.1 a_0), but not following it dwarf by dwarf (slope 0.32).

- The measured exponent is compatible with 3/2, halfway between 1 and 2 (their arithmetic mean). The profile M ∝ r^(3/2) is the geometric mean of the profiles M ∝ r and M ∝ r². This is an observation, not a derivation.
- **Limits of the data.**
- Ultra-faint dispersions can be inflated by binaries and tides.
- Y★ is fixed.
- Cluster σ_0 values come from models.

4.10 What it gives

system	result
Saturn (Cassini)	$Q_2 \approx 10^{-30}$ s^{-2} (Galactic tide only); measurement $(1.6 \pm 1.8)\cdot 10^{-27}$
planets, comets on any orbit, S2	Newtonian (dominant compact body)
919 wide binaries (0.95–3.7 r_M, e ~ 0.7)	$\langle\eta\rangle = 1.27$ (QUMOND chain 1.32; Newton 1)
wide binaries, direction	on DR3, once the close-pair level is subtracted, compatible with zero (w < 0.024)
disc galaxies	radial acceleration relation unchanged; environment field in full (SPARC)
eccentric orbits inside a galaxy	+6 % on the mean reading for r_a/r_p = 5.7 (no point mass at the centre)
light around galaxies	curvature, no external field: χ^2 219.9 on 60 KiDS points
light around galaxies in denser environments	more signal at large radii (all against isolated); no R² growth among central galaxies
classical dwarf spheroidals	within ±0.15 dex
ultra-faint dwarfs	2.5–20 times above the prediction
globular clusters beyond 20 kpc	Newtonian (Palomar 14 +0.06 dex, NGC 2419 +0.03)
dwarfs: who carries the mirror	all 38 systems with a metallicity spread
dwarfs: how much	M_dyn ∝ r_½^1.54±0.11 (1.40 without gas-rich systems), intrinsic scatter 0.48 dex

5. Registered for Gaia DR4 (lettura_locale_dr4/; k5c_dr4/ for the instantaneous form; lettura_locale/direzione/)

5.1 The boost

- **The law in the simulations.** Each relative orbit is Keplerian with G_eff = ν(x_eff + e_N). The pipeline programs of MM 2.56 are copied with exact, listed substitutions (*.diff): only the force law and the paths change.
- **Library.** 200 000 orbits, seed 18, in 337 s. |dE/E| is $4.5\cdot 10^{-13}$ at the 99th percentile (each orbit is Keplerian); no unbound initial state.
- **Check of the copies.** With the six hypotheses of MM 2.56 the evaluation reproduces the published dry run exactly: D = +7.64, subsets +7.56 / +4.50 / +2.07.
- **Decision rule.** D_L = L(local reading) − L(Newton); D_L,k is the same with one replica removed.
- **Local reading favoured:** D_L ≥ +5 and every D_L,k ≥ +2.
- **Newton favoured:** D_L ≤ −5 and every D_L,k ≤ −2.
- **Otherwise:** inconclusive.
- **Not valid:** if a validity condition of MM 2.56 fails.
- **Informative:**
- F_L = L(local reading) − L(chain);
- the slope statistic S over the two halves of the low band.

5.2 Dry run on DR3 (data already seen, grade B*; the registered simulation series R0)

hypothesis	ΔlogL from the best	deviance control / low
local reading	**0**	29.9 / 21.9
MOND simple	−6.10	43.3 / 20.7
MOND standard	−6.93	27.7 / 38.0
QUMOND chain, $H_0 = 67.4$	−8.46	37.6 / 31.1
QUMOND chain, $H_0 = 73.04$	−10.28	44.8 / 27.6
series, $a_0 = 1.2 \cdot 10^{-10}$	−14.82	48.6 / 32.9
Newton	−16.10	36.8 / 47.2

- D_L = +16.10 (subsets +16.92 / +16.00 / +6.46): the rule would give «local reading favoured over Newton».
- F_L = +8.46 (subsets +9.35 / +11.50 / +4.39).
- S: median u in the narrow and wide halves of the low band is 0.331 and 0.358 for the reading, against 0.342 and 0.353 in the data and 0.339 and 0.358 for the chain.
- **The instantaneous form** (k5c_dr4/: the same composition without the inertial state) is registered too. On DR3: D_K = +5.31 (subsets +2.07 / +9.04 / +9.43), F = −2.33.

5.3 Simulation noise on DR3 (documenti/NOTA_RUMORE_DR3.md)

- **The control band.** The mean boost of the local reading in the close pairs is 0.07 %. Yet their deviance changes by tens of points with the random seed alone:
- at zero companions the twin fraction enters only the seed string;
- there the deviance of the local reading moves from 240.1 to 213.7 at identical physics, and that of K5c from 313.6 to 222.4;
- the MM 2.56 simulations, which share the seed in this case, are identical.
- **Over the grid.** Across the 30 grid points the difference local − Newton in the control band has mean −7.7 and spread 20.9, with 15 positive and 15 negative values.
- **What it means.** The control-band differences between hypotheses are simulation noise, not a boost of close pairs. The DR3 differences D_L and F_L, and D of MM 2.56, carry a seed and library noise that the subset rule does not measure.
- **The measurement** (registered protocol and programs in lettura_locale/rumore). The series:
- R1 and R2 use new seeds with the same libraries;
- B1 uses new libraries for Newton (seed 21) and the local reading (seed 22).

The checks:

- the program reproduces a registered simulation exactly, and on R0 it gives D = +7.636 and D_L = +16.095;
- all 80 runs ended without errors.

series	D = chain − Newton	D_L = local − Newton	F_L = local − chain
R0 (registered)	+7.64	+16.10	+8.46
R1	+12.11	+10.12	−1.99
R2	−0.85	+8.20	+9.04
B1	—	+9.64	—
mean	**+6.30**	**+11.01**	**+5.17**
standard deviation	**6.58**	**3.49**	**6.21**

- **Against the criterion written beforehand** (standard deviation above 2.5), the DR3 differences are not stable with respect to the seed with the registered rule (three replicas in one series). R0–R2 use the registered libraries; B1 adds new ones, so D_L has four values and D and F_L three.
- The local reading stays above +5 over Newton in all four series.
- The chain over Newton, the quantity of MM 2.56, would have given «inconclusive» with another series of seeds.
- The local reading and the chain are not separated in a stable way on DR3.

5.4 A rule over several series for DR4 (lettura_locale/rumore/AGGIUNTA_DR4_PIU_SERIE.md)

- **Series.**
- Local reading and Newton: seven series, R0–R4 with the registered libraries and B1–B2 with new ones.
- The chain: five series.
- **The rule.**
- Local reading favoured: the mean $\bar{D}_L \geq +5$ and $D_L \geq +2$ in every series.
- Newton favoured: $\bar{D}_L \leq -5$ and $D_L \leq -2$ in every series.
- Otherwise: inconclusive.
- **Reporting.** Both rules are reported (the single series R0 with its subsets, and the several-series rule); the outcome is read with the several-series rule. The MM 2.56 rule for the chain stays as registered there.
- **Dry run on DR3** with the existing series: $\bar{D}_L = +11.0$ (minimum +8.2), «local reading favoured»; $\bar{D} = +6.3$ (minimum −0.85) and $\bar{F}_L = +5.2$ (minimum −2.0), inconclusive.
- **Cost at DR4.** 190 simulations, about five hours.
- **Programs** (dr4_programmi/, registered):
- esegui_dr4_serie_b.py runs the 190 simulations and stops with an error if one fails or is missing;
- valuta_dr4_serie_b.py applies the rule after the MM 2.56 validity conditions (V1 data, V2 libraries, grid and data fingerprints, V4 fit), on every series. On DR3 all conditions hold.
- **Checks on DR3.**
- The copied forward program reproduces a registered MM 2.56 simulation exactly.
- The evaluation reproduces the noise-test numbers (mean +11.01, minimum +8.20, «local reading favoured»).
- The two new libraries (Newton seed 23, local reading seed 24) are already generated.

5.5 The anisotropy (lettura_locale/direzione: BOZZA_PROTOCOLLO_ANISOTROPIA_DR4.md and AGGIUNTA_1)

- **Model.** $v^2 = v_0^2 (1 + w\, q \cos^2\theta)$, with w measured in each of three wide bands: 5 000–10 000, 10 000–20 000 and 20 000–50 000 au. Projection uses the MM 2.56 forward-model orbits.
- **Consistency.** The three values must agree (χ^2 with 2 degrees of freedom, $p > 0.05$), and the common w must be positive.
- **Sample effects.** The level of the 300–2 000 au band, where the reading predicts zero, is subtracted as a constant.
- **The dipole** is the primary statistic between the two forms:
- read by the dominant star: positive, about 1.9 times the quadrupole;
- symmetric: zero.
- The close-pair dipole must be zero, and the Galactic north pole is the control direction.
- **Addendum 2** (lettura_locale/direzione/TRADUZIONI_E_AGGIUNTA_2_DR4.md). The reading predicts the symmetric form: a zero dipole in every band, with all the anisotropy in the quadrupole.
- A positive dipole of about 1.9 times the quadrupole would favour the alternative.
- On DR3 the dipole is $+0.0027 \pm 0.0046$ dex.
- **Code** (dr4_programmi/anisotropia_dr4.py), registered before the DR4 release.
- It reads the files written by the MM 2.56 conversion.
- Dry run on DR3: S_ORB = 0.166 dex as in August; the close pairs show B2 = $+0.0037 \pm 0.0015$ and B1 = 0.0000 ± 0.0015.
- Outcome of the dry run: w compatible with zero in both forms (limits 0.024 and 0.014).

5.6 On 2 December 2026

The data are prepared with the steps of MM 2.56. Then:

- the 60 simulations of MM 2.56;
- the 10 + 10 simulations of the two local forms;
- valuta_lettura.py and valuta_k5c_c1.py in dr4 mode;
- esegui_dr4_serie_b.py <DR4 data> <output> tutto, then valuta_dr4_serie_b.py <DR4 data> <output> dr4 (section 5.4);
- anisotropia_dr4.py <DR4 data> <output> dr4 (section 5.5).

6. Predictions the data will read

- **Gaia DR4:**
- the boost of wide binaries and its rise with separation;
- the anisotropy: positive, only in wide pairs, growing as $q \propto s^2/M$, with the same w in every band and no change of sign (full QUMOND changes sign near 12 000 au);
- a zero dipole (symmetric form); a positive one would favour the reading by the dominant star;
- no significant dependence of the boost on the mass ratio below the prevalence threshold, on many more unequal pairs.
- **Systems without rotation:**
- distant globular clusters Newtonian;
- new dwarfs with stars of different generations on the same profile, M_dyn ∝ r_½^1.5;
- new single-generation clusters Newtonian.
- **Light companions.** A companion much lighter than its star at a wide separation (a brown dwarf or a planet at 10^4 au) is read by the star itself and is Newtonian. Pairs of comparable masses are boosted. Between mass ratios 1/1 048 and 1/7.4 the threshold will be set by the data.
- **Comets and the Oort cloud:** Newtonian, on any orbit.
- **Weak lensing:** no external-field effect around galaxies, in any environment; the dynamics of stars and binaries does show it.

7. What this record does not contain

- A covariant field theory of the local reading: here it is a law of motion for bound relations, with a reading of light by curvature.
- The direction term inside the DR4 simulations of the boost (its mean effect is +0.002 dex).
- A derivation of the common profile of the dwarfs (its exponent and normalisation are measured).
- Any change to the MM 2.56 protocol.

8. Verification

- Codex/GPT-6 checked Levels 1–2 (documenti/verifica_astra/RAPPORTO_GIRO1.md) and the instantaneous DR4 registration (k5c_dr4/documenti/verifica_astra/RAPPORTO_K5C.md).
- A third round checked the full local reading (lettura_locale_dr4/documenti/verifica_astra/RAPPORTO_LETTURA.md). It confirmed the dry-run numbers and the exact reproduction of MM 2.56.
- Changes of wording and documentation were applied (the CORREZIONI_* files); no arithmetic error changing an outcome was found.
- A fourth round checked the additions after the second version: sections 4.3–4.9 and 5.3–5.5 (documenti/verifica_astra/RAPPORTO_PARTI_NUOVE.md).
- Two remarks did not hold: a formatting error in dipolo_dominante.py (tuple addition concatenates) and an unchecked star order (star 1 is the brighter in all rows of both catalogues).
- The others were applied:
- the w fit with the full covariance;
- numeric rules for the anisotropy protocol (addendum 3);
- the Y★ sensitivity and a comparison at the same gravitational state for dwarfs and clusters;
- wording on S2, the fast pairs, the lenses and the mass ratio;
- a recount of the SPARC concentration.
- No outcome changed (documenti/CORREZIONI_DOPO_ASTRA_V3.md).
- A fifth round checked the additions after the fourth: the total field, the red and blue lenses, dwarfs and clusters, and the DR4 programs (documenti/verifica_astra/RAPPORTO_QUINTO_GIRO.md).
- Applied corrections (documenti/CORREZIONI_DOPO_ASTRA_V4.md):
- the profile fit with the correct radius-error model;
- the partial correlation as Pearson on rank residuals;
- the dwarf comparison without gas-rich systems;
- P3 without the Magellanic Clouds;
- validity and completeness checks in the DR4 programs (versions _b);
- softened causal wording (generations, tide, satellites);
- «3/2 is the arithmetic mean of 1 and 2».
- Remarks checked and not applied: the meaning of rhalf_sph_physical (our M_dyn equals the catalogue's Wolf mass within 0.6 %) and the counts of measured metallicity spreads (38, as written).
- No outcome changed.
- Every protocol and program was registered with SHA-256 and UTC time before it was run (the registers are included).

9. Files and reproducibility

- `codice/` and `risultati/`: Level 1 (passo123.py, passo4.py) and Level 2 (livello2.py, livello2_c1.py), with logs and JSON. `codice/esterni/`: the earlier verification code and the MM 2.56 solver, unchanged.
- `lettura_locale/`: the checks of section 4, with logs and JSON:
- premessa_mcd.py and premessa_inerzia.py;
- corpo_dominante/, coppie/, campo/ (with the 10 simulations of the total-field run), direzione/, luce/, galassie/ and sistemi/;
- rumore/ (sections 5.3–5.4), with the 80 simulations of the noise test and their SHA-256.

REGISTRO_estratto.txt has the registration lines. Some files differ from their registered originals only in the text lines listed in CAMBI_TESTO.md: five programs, the noise protocol and addendum 2. The code fingerprints are identical.

- `lettura_locale_dr4/` and `k5c_dr4/`: the DR4 registrations (protocols, copy generators and diffs, programs, logs, the 10 + 10 DR3 simulations with SHA-256). The orbit libraries (about 210 MB each) are not included; they regenerate in about six minutes from the seeds 18 and 17.
- `dr4_programmi/`: the registered DR4 programs (anisotropy; the rule over several series), their dry runs on DR3, the identity check and the logs of the two new libraries. The libraries are not included; they regenerate from seeds 23 and 24.
- `dati/`: the MM 2.56 sample, the JPL page, the source list.
- `documenti/`: inventories, protocols, outcome notes, the structure of the local reading and its checks, the note on simulation noise, verification reports, registers.
- The programs contain the author's paths (E:\...), to be adapted.

The author is not a professional physicist. Computations were done with AI assistants (Claude, with checks by Codex), not reviewed by a human referee. Texts and materials: CC BY 4.0; code: MIT.

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